WORKED SOLUTION

STEP II 2008, Question 13

Probability and Statistics • Drawing Without Replacement

STEP II 2008, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s2-q13

The key idea

Don't track every counter: track only the black ones. Each black counter's fate is decided by two draws, and at each draw the chance that a particular counter is taken is simply (number drawn)/(number in the bag). Splitting into cases according to what happens at the first draw turns both parts into short sums, and both simplify far more than you might expect.

Solution

Throughout, "drawn at random" means every set of k counters in the bag is equally likely to be the one drawn. The fact we use repeatedly is this: if k counters are drawn at random from a bag of m, a *particular* counter is among them with probability k/m, by symmetry (each of the m counters is equally likely to be included, and on average k of them are).

After the first transfer, bag P holds $n-k$ counters and bag Q holds $n+k$. At the second transfer, k of those $n+k$ are drawn and moved to P.

Part (i)

Let B be the black counter, starting in P. Split according to whether B moves at the first transfer.

- B is not drawn from P: probability $\dfrac{n-k}{n}$. Then B stays in P throughout, since the second transfer only takes counters out of Q.
- B is drawn from P: probability $\dfrac{k}{n}$. Now B is one of the $n+k$ counters in Q, and it returns to P exactly when it is among the k drawn back, which has probability $\dfrac{k}{n+k}$.

These cases are disjoint and cover everything, so

$$\begin{aligned}\mathrm{P}(B \text{ ends in } P) &= \frac{n-k}{n} + \frac{k}{n}\cdot\frac{k}{n+k} = \frac{(n-k)(n+k)+k^2}{n(n+k)}\\ &= \frac{n^2}{n(n+k)} = \frac{n}{n+k}.\end{aligned}$$

This is strictly decreasing in k, so over $0 \le k \le n$ it is maximised at $k = 0$ only, where it equals 1. (Not surprising: if nothing is moved, the black counter certainly stays put.)

Part (ii)

Now let B_1 be the black counter starting in P and B_2 the one starting in Q. Let S be the event that they end in the same bag. Again split on whether B_1 moves at the first transfer.

Case 1: B_1 stays in P (probability $\frac{n-k}{n}$). Then B_1 ends in P whatever happens, so S happens exactly when B_2 is among the k drawn from the $n+k$ counters in Q:

$$\mathrm{P}(S \mid \text{case 1}) = \frac{k}{n+k}.$$

Case 2: B_1 moves to Q (probability $\frac{k}{n}$). Now both black counters are among the $n+k$ in Q, and S happens when the second draw takes *both* of them or *neither*.

> **Ask yourself** What is the probability that a random k-set from $n+k$ counters contains both, or neither, of two given counters?

Count subsets. There are $\binom{n+k}{k}$ equally likely choices. Those containing both blacks choose the other $k-2$ from the remaining $n+k-2$; those containing neither choose all k from the $n+k-2$ whites. Dividing,

$$\begin{aligned}\mathrm{P}(\text{both}) &= \binom{n+k-2}{k-2}\Big/\binom{n+k}{k} = \frac{k(k-1)}{(n+k)(n+k-1)},\\ \mathrm{P}(\text{neither}) &= \binom{n+k-2}{k}\Big/\binom{n+k}{k} = \frac{n(n-1)}{(n+k)(n+k-1)}.\end{aligned}$$

(These formulas also hold for $k = 0, 1$, where $\mathrm{P}(\text{both}) = 0$.) So

$$\mathrm{P}(S \mid \text{case 2}) = \frac{k(k-1)+n(n-1)}{(n+k)(n+k-1)}.$$

Putting the cases together. By the law of total probability,

$$\begin{aligned}\mathrm{P}(S) &= \frac{n-k}{n}\cdot\frac{k}{n+k} + \frac{k}{n}\cdot\frac{k(k-1)+n(n-1)}{(n+k)(n+k-1)}\\ &= \frac{k\big[(n-k)(n+k-1)+k^2-k+n^2-n\big]}{n(n+k)(n+k-1)}.\end{aligned}$$

Inside the square bracket, $(n-k)(n+k-1) = n^2 - k^2 - n + k$, so the bracket is $2n^2 - 2n = 2n(n-1)$. Hence

$$\boxed{\mathrm{P}(S) = \frac{2k(n-1)}{(n+k)(n+k-1)}}$$

Maximising. At $k = 0$ the probability is 0, so take $1 \le k \le n$ and minimise the reciprocal. Up to the positive constant $2(n-1)$,

$$\frac{(n+k)(n+k-1)}{k} = k + (2n-1) + \frac{n(n-1)}{k},$$

so we need to minimise $g(k) = k + \dfrac{n(n-1)}{k}$ over integers. Compare neighbours:

$$g(k+1) - g(k) = 1 - \frac{n(n-1)}{k(k+1)}.$$

This is negative when $k(k+1) < n(n-1)$, that is for $k < n-1$, and it is exactly zero when $k = n-1$, because then $k(k+1) = (n-1)n$. So g strictly decreases from $k = 1$ to $k = n-1$, and $g(n) = g(n-1)$. Therefore the probability is maximised at

$$k = n-1 \quad \text{and} \quad k = n,$$

both of which lie in the allowed range since $n \ge 2$. At $k = n-1$ the maximum value is

$$\frac{2(n-1)^2}{(2n-1)(2n-2)} = \frac{n-1}{2n-1},$$

and $k = n$ gives the same, $\dfrac{2n(n-1)}{2n(2n-1)} = \dfrac{n-1}{2n-1}$. (The continuous minimiser of g is $\sqrt{n(n-1)}$, which lies strictly between $n-1$ and n; the tie between the two integers is exact, not approximate.)

Checks

- Part (i) at $k = n$: $\frac{1}{2}$. Then all $2n$ counters pass through Q and n of them return, so the black one is equally likely to end in either bag. Good.
- Part (ii) at $k = 0$ gives 0, correct since the blacks never meet. At $n = 2$, $k = 1$ it gives $\frac{2}{3 \cdot 2} = \frac{1}{3}$, which you can confirm by hand.
- We enumerated every possible pair of draws exactly (with fractions) for $2 \le n \le 7$ and every k; both formulas matched in every case, and the maximisers in (ii) were always $\{n-1, n\}$, e.g. maximum $\frac{3}{7} \approx 0.4286$ for $n = 4$.

Common slips

- Using k/n instead of $k/(n+k)$ at the second draw: bag Q has grown to $n+k$ counters.

- In part (ii), forgetting the "neither drawn" possibility in case 2: if both blacks stay in Q they are still together.
- Maximising by calculus and rounding $\sqrt{n(n-1)}$ to a single integer. The discrete comparison shows two equal maximisers.

Beyond STEP

The tool behind both parts is the *hypergeometric* distribution: the number of marked items in a random sample drawn without replacement. Conditioning on the outcome of the first stage and then using it for the second is the law of total probability, and the same two-stage reasoning underlies models of mixing and diffusion between containers, such as the Ehrenfest urn. Blitzstein and Hwang, *Introduction to Probability*, treats the hypergeometric distribution and conditioning in exactly this style.

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